Homework 3: Frequency Response

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Class: EE 522 Discrete Time Processing

AI Disclosure:

AI was used in the following ways:

- Code generation
- Markdown formatting
- Derivation assistance

How $y[n]$ and $H(e^{j\omega})$ are computed

For an LTI system with impulse response $h[n]$, the output is the convolution

$$y[n] = (h * x)[n] = \sum_{k=-\infty}^{\infty} h[k]\, x[n-k]$$

Step response ($x[n] = u[n]$): since $u[n-k] = 1$ only when $k \le n$,

$$s[n] = \sum_{k=-\infty}^{n} h[k], \qquad \lim_{n\to\infty} s[n] = \sum_{k=-\infty}^{\infty} h[k] = H(e^{j0})$$

Frequency response (the DTFT of $h[n]$):

$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h[n]\, e^{-j\omega n} = |H(e^{j\omega})|\, e^{j\angle H(e^{j\omega})} = \mathrm{Re}\{H\} + j\,\mathrm{Im}\{H\}$$

$H(e^{j\omega})$ is 2π-periodic, which is why every frequency plot repeats three times over $-3\pi \le \omega \le 3\pi$. For real $h[n]$, $|H|$ and $\mathrm{Re}\{H\}$ are even and $\angle H$ and $\mathrm{Im}\{H\}$ are odd. Plotted phase is wrapped into $(-\pi, \pi]$.

Identities used below:

$$\sum_{n=0}^{M-1} a^n = \frac{1-a^M}{1-a}, \qquad 1 - e^{-j\theta} = 2j\sin(\theta/2)\, e^{-j\theta/2}, \qquad 1 + e^{-j\theta} = 2\cos(\theta/2)\, e^{-j\theta/2}$$

Problem 1: $h[n] = u[n] - u[n-7]$

Derivation

Step 1: Write out $h[n]$ sample by sample.

$u[n] = 1$ for $n \geq 0$, and $u[n-7] = 1$ for $n \geq 7$. Subtracting removes everything from $n = 7$ onward, which leaves ones only for $0 \leq n \leq 6$:

$$h[n] = u[n] - u[n-7] = \begin{cases} 1, & 0 \leq n \leq 6 \\ 0, & \text{otherwise} \end{cases}$$

n	≤ -1	0	1	2	3	4	5	6	≥ 7
$h[n]$	0	1	1	1	1	1	1	1	0

This is seven ones starting at $n = 0$. Its center of symmetry is $n = 3$. *(Impulse-response plot.)*

Step 2: Output for any input $x[n]$.

Start from the convolution sum:

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]\, x[n-k]$$

$h[k] = 1$ only for $k = 0, \ldots, 6$, so only those seven terms survive:

$$y[n] = \sum_{k=0}^{6} x[n-k] = x[n] + x[n-1] + x[n-2] + x[n-3] + x[n-4] + x[n-5] + x[n-6]$$

Each output is the sum of the current input and the 6 inputs before it, a 7-point moving sum. It uses only present and past inputs. *(Random-input plot: the output is a smoothed, larger version of the input, and it continues until $n = 99 + 6 = 105$.)*

Step 3: Step response, $x[n] = u[n]$.

Substitute $x[n] = u[n]$ into Step 2:

$$s[n] = \sum_{k=0}^{6} u[n-k]$$

$u[n-k] = 1$ only when $k \le n$, so $s[n]$ counts how many of $k = 0, \dots, 6$ satisfy $k \le n$:

- $n < 0$: no values of k qualify, so $s[n] = 0$
- $0 \le n \le 6$: $k = 0, 1, \dots, n$ qualify, which is $n + 1$ values
- $n > 6$: all 7 values qualify, so $s[n] = 7$

$$s[n] = \begin{cases} 0, & n < 0 \\ n+1, & 0 \le n \le 6 \\ 7, & n > 6 \end{cases}$$

For example, $s[0] = 1$, $s[3] = 4$, and $s[6] = 7$. *(Step-response plot: a ramp from $n = 0$ that levels off at 7.)*

Step 4: Frequency response, written out term by term.

Start from the DTFT definition. Only the seven nonzero samples contribute:

$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h[n]\, e^{-j\omega n} = \sum_{n=0}^{6} e^{-j\omega n} = 1 + e^{-j\omega} + e^{-j2\omega} + e^{-j3\omega} + e^{-j4\omega} + e^{-j5\omega} + e^{-j6\omega}$$

Use $e^{-j\theta} = \cos\theta - j\sin\theta$ on every term to split real and imaginary parts:

$$\begin{aligned} \mathrm{Re}\{H\} &= 1 + \cos\omega + \cos 2\omega + \cdots + \cos 6\omega, \\ \mathrm{Im}\{H\} &= -\left(\sin\omega + \sin 2\omega + \cdots + \sin 6\omega\right) \end{aligned}$$

Unlike Problem 2, the sine terms don't cancel, because $h[n]$ is not symmetric about $n = 0$. So H is complex.

Step 5: Closed form (explains the shape of the plots).

Use the finite geometric series $\sum_{m=0}^{M-1} a^m = \dfrac{1 - a^M}{1 - a}$ with $a = e^{-j\omega}$ and $M = 7$:

$$H(e^{j\omega}) = \sum_{n=0}^{6} \left(e^{-j\omega}\right)^n = \frac{1 - e^{-j7\omega}}{1 - e^{-j\omega}}$$

Factor half of each exponent out of the numerator and denominator, then use $e^{j\theta} - e^{-j\theta} = 2j\sin\theta$:

$$1 - e^{-j7\omega} = e^{-j7\omega/2}\left(e^{j7\omega/2} - e^{-j7\omega/2}\right) = e^{-j7\omega/2} \cdot 2j\sin(7\omega/2)$$

$$1 - e^{-j\omega} = e^{-j\omega/2}\left(e^{j\omega/2} - e^{-j\omega/2}\right) = e^{-j\omega/2} \cdot 2j\sin(\omega/2)$$

Divide. The $2j$ cancels, and $e^{-j7\omega/2} / e^{-j\omega/2} = e^{-j3\omega}$:

$$\boxed{H(e^{j\omega}) = e^{-j3\omega}\,\frac{\sin(7\omega/2)}{\sin(\omega/2)}}$$

The answer has two parts: a real amplitude, $\dfrac{\sin(7\omega/2)}{\sin(\omega/2)}$, which is exactly Problem 2's H, multiplied by a pure phase factor $e^{-j3\omega}$ that comes from the 3-sample delay to the center of $h[n]$.

Step 6: Magnitude, phase, real, and imaginary parts.

Let $A(\omega) = \dfrac{\sin(7\omega/2)}{\sin(\omega/2)}$ (real). Since $|e^{-j3\omega}| = 1$:

$$|H(e^{j\omega})| = |A(\omega)| = \left|\frac{\sin(7\omega/2)}{\sin(\omega/2)}\right|$$

$$\angle H(e^{j\omega}) = \begin{cases} -3\omega, & A(\omega) > 0 \\ -3\omega + \pi, & A(\omega) < 0 \end{cases} \quad (\text{then wrapped into } (-\pi, \pi])$$

Expand $e^{-j3\omega} = \cos 3\omega - j\sin 3\omega$:

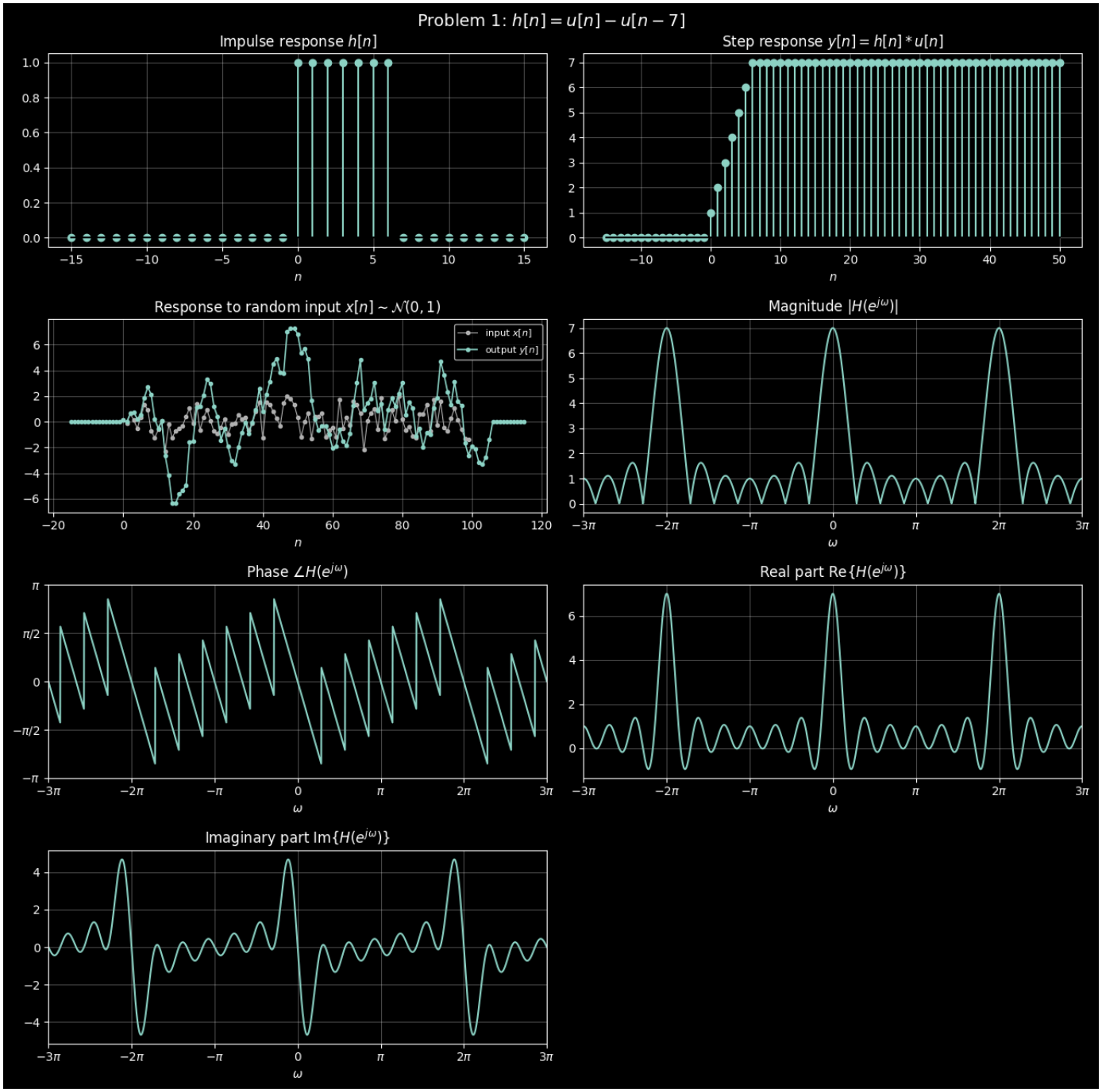
$$\mathrm{Re}\{H\} = \cos(3\omega)\,A(\omega), \qquad \mathrm{Im}\{H\} = -\sin(3\omega)\,A(\omega)$$

The phase is a straight line with slope -3, but wrapping it into $(-\pi, \pi]$ turns it into the steep sawtooth in the plot, with extra jumps of π wherever $A(\omega)$ changes sign.

Step 7: Key values to check against the plots.

Quantity	Value	Where to see it
$H(e^{j0})$	$1+1+\cdots+1$ $= 7$	Peak of magnitude and real part; final value of the step response
$H(e^{j\pi})$	$1-1+1-1+1$ $-1+1=1$	Magnitude equals 1 at $\omega = \pm\pi$
Zeros of H	$\sin(7\omega/2) = 0$ $\Rightarrow \omega = \pm\frac{2\pi}{7}$, $\pm\frac{4\pi}{7}, \pm\frac{6\pi}{7}$	Magnitude touches 0 six times per period, the same as Problem 2
Phase slope	-3 (a 3-sample delay)	Sawtooth in the phase plot
$\mathrm{Im}\{H\}$	Odd; 0 at $\omega = 0$; extremes $\approx \pm 4.7$	Imaginary-part plot
Period	2π	Every frequency plot repeats at $\omega = \pm 2\pi$

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Classification

Property	Answer	Justification
Filter type	Lowpass	When I compare $\lvert H(e^{j\omega})\rvert$ at ω = 0 (DC) and ω = π, the peak is at ω = 0 and is small at ω = π
Causality	Causal	$h[n] = 0$ for all n < 0
Stability	Stable	$\Sigma \lvert h[n] \rvert < \infty$
FIR or IIR	FIR	There are only 7 nonzero samples

Problem 2: $h[n] = u[n+3] - u[n-4]$

Derivation

Step 1: Write out $h[n]$ sample by sample.

$u[n+3] = 1$ for $n \geq -3$, and $u[n-4] = 1$ for $n \geq 4$. Subtracting removes everything from $n = 4$ onward, which leaves ones only for $-3 \leq n \leq 3$:

$$h[n] = u[n+3] - u[n-4] = \begin{cases} 1, & -3 \leq n \leq 3 \\ 0, & \text{otherwise} \end{cases}$$

n	≤ -4	-3	-2	-1	0	1	2	3	≥ 4
$h[n]$	0	1	1	1	1	1	1	1	0

This is seven ones centered on $n = 0$, the same as Problem 1 shifted 3 samples to the left. *(Impulse-response plot.)*

Step 2: Output for any input $x[n]$.

Start from the convolution sum:

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]\, x[n-k]$$

$h[k] = 1$ only for $k = -3, \ldots, 3$, so only those seven terms survive:

$$y[n] = \sum_{k=-3}^{3} x[n-k] = x[n+3] + x[n+2] + x[n+1] + x[n] + x[n-1] + x[n-2] + x[n-3]$$

Each output is the sum of the current input, the 3 inputs before it, and the 3 inputs after it. Because it needs future inputs ($x[n+1]$, $x[n+2]$, $x[n+3]$), the system is non-causal. *(Random-input plot: the output starts at $n = -3$, before the input starts at $n = 0$, and is a smoothed version of the input.)*

Step 3: Step response, $x[n] = u[n]$.

Substitute $x[n] = u[n]$ into Step 2:

$$s[n] = \sum_{k=-3}^{3} u[n-k]$$

$u[n-k] = 1$ only when $k \le n$, so $s[n]$ counts how many of $k = -3, \ldots, 3$ satisfy $k \le n$:

- $n < -3$: no values of k qualify, so $s[n] = 0$
- $-3 \le n \le 3$: $k = -3, -2, \ldots, n$ qualify, which is $n - (-3) + 1 = n + 4$ values
- $n > 3$: all 7 values qualify, so $s[n] = 7$

$$s[n] = \begin{cases} 0, & n < -3 \\ n + 4, & -3 \le n \le 3 \\ 7, & n > 3 \end{cases}$$

For example, $s[-3] = 1$, $s[0] = 4$, and $s[3] = 7$. *(Step-response plot: a ramp from $n = -3$ that levels off at 7.)*

Step 4: Frequency response, written out term by term.

Start from the DTFT definition. Only the seven nonzero samples contribute:

$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h[n]\, e^{-j\omega n} = \sum_{n=-3}^{3} e^{-j\omega n}$$

Write out all seven terms:

$$H(e^{j\omega}) = e^{j3\omega} + e^{j2\omega} + e^{j\omega} + 1 + e^{-j\omega} + e^{-j2\omega} + e^{-j3\omega}$$

Group each positive exponent with its negative partner:

$$H(e^{j\omega}) = 1 + \left(e^{j\omega} + e^{-j\omega}\right) + \left(e^{j2\omega} + e^{-j2\omega}\right) + \left(e^{j3\omega} + e^{-j3\omega}\right)$$

Apply Euler's formula, $e^{j\theta} + e^{-j\theta} = 2\cos\theta$, to each pair:

$$\boxed{H(e^{j\omega}) = 1 + 2\cos\omega + 2\cos 2\omega + 2\cos 3\omega}$$

Every term is a real cosine, so $H(e^{j\omega})$ is **purely real**. This happens because $h[n]$ is symmetric about $n = 0$ ($h[n] = h[-n]$).

Step 5: Closed form (explains the shape of the plots).

Re-index with $m = n + 3$ so the sum starts at 0, then use the finite geometric series $\sum_{m=0}^{M-1} a^m = \dfrac{1 - a^M}{1 - a}$ with $a = e^{-j\omega}$ and $M = 7$:

$$H(e^{j\omega}) = \sum_{m=0}^{6} e^{-j\omega(m-3)} = e^{j3\omega} \sum_{m=0}^{6} e^{-j\omega m} = e^{j3\omega}\, \frac{1 - e^{-j7\omega}}{1 - e^{-j\omega}}$$

Factor half of each exponent out of the numerator and denominator, then use $e^{j\theta} - e^{-j\theta} = 2j\sin\theta$:

$$1 - e^{-j7\omega} = e^{-j7\omega/2}\left(e^{j7\omega/2} - e^{-j7\omega/2}\right) = e^{-j7\omega/2} \cdot 2j\sin(7\omega/2)$$
$$1 - e^{-j\omega} = e^{-j\omega/2}\left(e^{j\omega/2} - e^{-j\omega/2}\right) = e^{-j\omega/2} \cdot 2j\sin(\omega/2)$$

Divide. The $2j$ cancels, and $e^{-j7\omega/2}/e^{-j\omega/2} = e^{-j3\omega}$ cancels the $e^{j3\omega}$ in front:

$$\boxed{H(e^{j\omega}) = \frac{\sin(7\omega/2)}{\sin(\omega/2)}}$$

At $\omega = 0$ this is $0/0$, but Step 4 gives the value directly: $H(e^{j0}) = 1 + 2 + 2 + 2 = 7$.

Step 6: Magnitude, phase, real, and imaginary parts.

Because H is real:

$$\mathrm{Re}\{H(e^{j\omega})\} = \frac{\sin(7\omega/2)}{\sin(\omega/2)}, \qquad \mathrm{Im}\{H(e^{j\omega})\} = 0$$

$$|H(e^{j\omega})| = \left|\frac{\sin(7\omega/2)}{\sin(\omega/2)}\right|, \qquad \angle H(e^{j\omega}) = \begin{cases} 0, & H(e^{j\omega}) > 0 \\ \pi, & H(e^{j\omega}) < 0 \end{cases}$$

The angle of a real number is 0 if it's positive and π if it's negative, which is why the phase plot is a square wave.

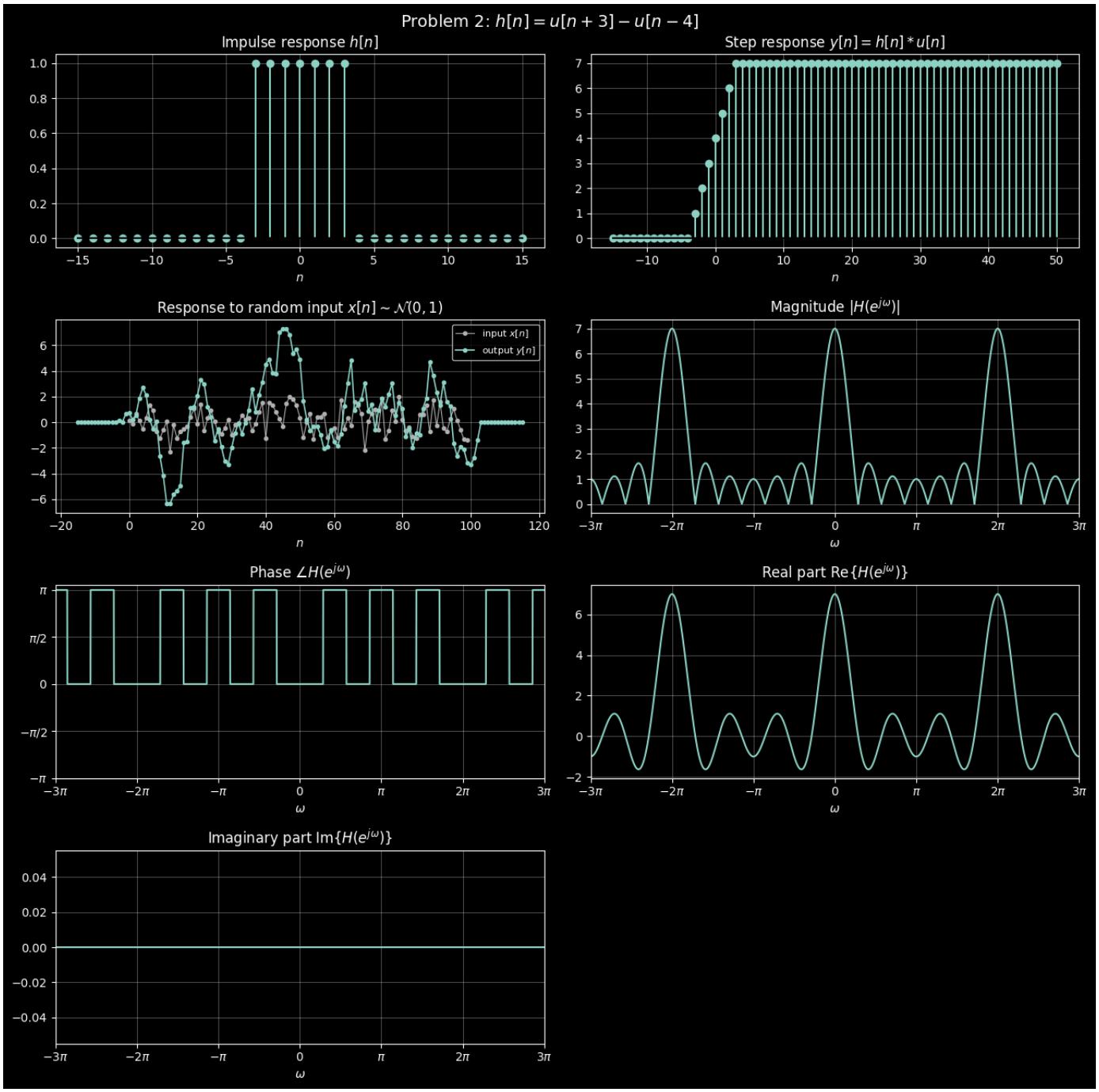
Step 7: Key values to check against the plots.

Quantity	Value	Where to see it
$H(e^{j0})$	$1 + 2 + 2 + 2 = 7$	Peak of magnitude and real part; final value of the step response
$H(e^{j\pi})$	$1 - 2 + 2 - 2 = -1$	Real part equals -1 at $\omega = \pm\pi$; phase equals π there
Zeros of H	$\sin(7\omega/2) = 0 \Rightarrow \omega = \pm\frac{2\pi}{7}, \pm\frac{4\pi}{7}, \pm\frac{6\pi}{7}$	Magnitude touches 0 six times per period
Sign of H on $0 < \omega < \pi$	$+$ on $(0, \frac{2\pi}{7})$, $-$ on $(\frac{2\pi}{7}, \frac{4\pi}{7})$, $+$ on $(\frac{4\pi}{7}, \frac{6\pi}{7})$, $-$ on $(\frac{6\pi}{7}, \pi)$	Phase switches between 0 and π at each zero
Period	2π	Every frequency plot repeats at $\omega = \pm 2\pi$

Comparison with Problem 1: $h_2[n] = h_1[n+3]$, and a shift of 3 samples to the left multiplies the DTFT by $e^{j3\omega}$. That factor has magnitude 1, so $|H|$ is identical to Problem 1's. It also cancels Problem

1's $e^{-j3\omega}$ linear phase, which is why H here is purely real.

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ba check: 0.0
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Classification

Property	Answer	Justification
Filter type	Lowpass	When I compare $\left\|H(e^{j\omega})\right\|$ at ω = 0 (DC) and ω = π, the peak is at ω = 0 and is small at ω = π
Causality	Non-causal	There are samples such that $h[n] \neq 0$ for $n < 0$

Property	Answer	Justification
Stability	Stable	$\Sigma\lvert h[n]\rvert = 7 < \infty$
FIR or IIR	FIR	There are only 7 nonzero samples

Problem 3: $h[n] = \delta[n] - \delta[n-1] + \delta[n-2] - \delta[n-3]$

Derivation

Step 1: Write out $h[n]$ sample by sample.

Each $\delta[n-m]$ places a single sample at $n = m$, with the sign in front of it:

$$h[n] = \delta[n] - \delta[n-1] + \delta[n-2] - \delta[n-3]$$

n	≤ -1	0	1	2	3	≥ 4
$h[n]$	0	1	-1	1	-1	0

Four taps with alternating signs. *(Impulse-response plot.)*

Step 2: Output for any input $x[n]$.

In the convolution sum $y[n] = \sum_k h[k]\, x[n-k]$, only $k = 0, 1, 2, 3$ survive, each weighted by its tap:

$$y[n] = (1)\, x[n] + (-1)\, x[n-1] + (1)\, x[n-2] + (-1)\, x[n-3] = x[n] - x[n-1] + x[n-2] - x[n-3]$$

Slowly changing inputs nearly cancel, since $+x - x + x - x \approx 0$, while inputs that alternate in sign add up. *(Random-input plot: the output is rougher than the input. Its variance is $1^2 + 1^2 + 1^2 + 1^2 = 4$, so its standard deviation is about 2.)*

Step 3: Step response, $x[n] = u[n]$.

The step response is the running sum of the taps, $s[n] = \sum_{k=-\infty}^{n} h[k]$:

n	≤ -1	0	1	2	≥ 3
running sum	0	1	$1 - 1 = 0$	$0 + 1 = 1$	$1 - 1 = 0$

$$s[n] = \delta[n] + \delta[n-2]$$

The final value is 0 because the taps add up to 0. A constant (DC) input produces no output. *(Step-response plot: two unit spikes at $n = 0$ and $n = 2$, then 0.)*

Step 4: Frequency response, written out term by term.

Each tap $h[n]$ contributes $h[n]\,e^{-j\omega n}$:

$$H(e^{j\omega}) = \sum_{n=0}^{3} h[n]\,e^{-j\omega n} = 1 - e^{-j\omega} + e^{-j2\omega} - e^{-j3\omega}$$

Use $e^{-j\theta} = \cos\theta - j\sin\theta$ on every term to split real and imaginary parts:

$$\mathrm{Re}\{H\} = 1 - \cos\omega + \cos 2\omega - \cos 3\omega, \qquad \mathrm{Im}\{H\} = \sin\omega - \sin 2\omega + \sin 3\omega$$

Step 5: Closed form by factoring (explains the shape of the plots).

Group the four terms in pairs and pull out the common factor:

$$H(e^{j\omega}) = \left(1 - e^{-j\omega}\right) + e^{-j2\omega}\left(1 - e^{-j\omega}\right) = \left(1 - e^{-j\omega}\right)\left(1 + e^{-j2\omega}\right)$$

Simplify each factor by pulling out half of its exponent:

$$1 - e^{-j\omega} = e^{-j\omega/2}\left(e^{j\omega/2} - e^{-j\omega/2}\right) = 2j\sin(\omega/2)\,e^{-j\omega/2}$$
$$1 + e^{-j2\omega} = e^{-j\omega}\left(e^{j\omega} + e^{-j\omega}\right) = 2\cos(\omega)\,e^{-j\omega}$$

Multiply them back together:

$$\boxed{H(e^{j\omega}) = 4j\,\sin(\omega/2)\cos(\omega)\,e^{-j3\omega/2}}$$

Each factor accounts for part of the magnitude plot: $\sin(\omega/2)$ creates the zero at $\omega = 0$, and $\cos\omega$ creates the zeros at $\omega = \pm\pi/2$.

Step 6: Magnitude, phase, real, and imaginary parts.

Write $j = e^{j\pi/2}$ so all the phase terms are together:

$$H(e^{j\omega}) = \underbrace{4\sin(\omega/2)\cos\omega}_{\text{real amplitude}}\; e^{\,j\left(\frac{\pi}{2} - \frac{3\omega}{2}\right)}$$

$$|H(e^{j\omega})| = 4\,|\sin(\omega/2)\cos\omega|$$

$$\angle H(e^{j\omega}) = \begin{cases} \dfrac{\pi}{2} - \dfrac{3\omega}{2}, & \sin(\omega/2)\cos\omega > 0 \\ \dfrac{\pi}{2} - \dfrac{3\omega}{2} + \pi, & \sin(\omega/2)\cos\omega < 0 \end{cases} \qquad (\text{then wrapped into } (-\pi, \pi])$$

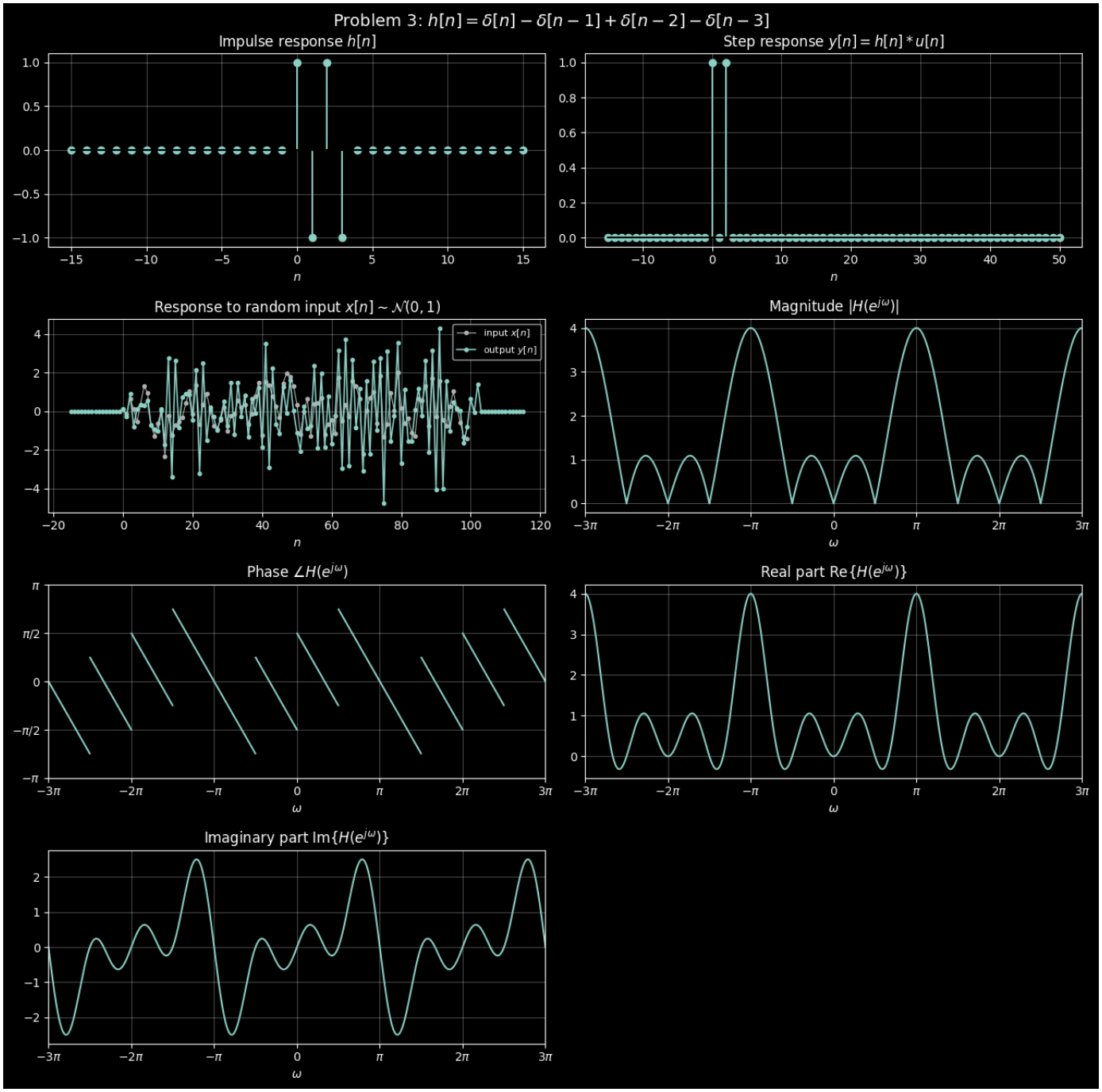
The real and imaginary parts are the ones from Step 4. The phase has a slope of $-3/2$, a 1.5-sample delay to the center of $h[n]$, plus a constant $\pi/2$ because $h[n]$ is antisymmetric about

$n = 1.5$ ($h[0] = -h[3]$, $h[1] = -h[2]$).

Step 7: Key values to check against the plots.

Quantity	Value	Where to see it
$H(e^{j0})$	$1 - 1 + 1 - 1 = 0$	Magnitude is 0 at DC; step response ends at 0
$H(e^{j\pi})$	$1 + 1 + 1 + 1 = 4$	Peak of magnitude and real part at $\omega = \pm\pi$
Zeros of H	$\omega = 0$ and $\cos\omega = 0 \Rightarrow \omega = \pm\frac{\pi}{2}$	Magnitude touches 0 three times per period
Small humps	$\lvert H\rvert \approx 1.08$ near $\omega = \pm\frac{\pi}{4}$	Between the zeros at 0 and $\pm\frac{\pi}{2}$
Phase slope	$-3/2$ (1.5-sample delay)	Descending segments in the phase plot
Period	2π	Every frequency plot repeats at $\omega = \pm 2\pi$

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Classification

Property	Answer	Justification
Filter type	Highpass	When I compare $\lvert H(e^{j\omega})\rvert$ at ω = 0 (DC) and ω = π, the peak is at ω = π and most minimum at ω = 0
Causality	Causal	$h[n] = 0$ for all n < 0
Stability	Stable	$\Sigma\lvert h[n]\rvert = 1+1+1+1 = 4 < \infty$
FIR or IIR	FIR	There are only 4 nonzero samples.

Problem 4: $h[n] = \delta[n] + \delta[n-1] - \delta[n-2] - \delta[n-3]$

Derivation

Step 1: Write out $h[n]$ sample by sample.

$$h[n] = \delta[n] + \delta[n-1] - \delta[n-2] - \delta[n-3]$$

n	≤ -1	0	1	2	3	≥ 4
$h[n]$	0	1	1	-1	-1	0

Two positive taps followed by two negative taps. *(Impulse-response plot.)*

Step 2: Output for any input $x[n]$.

In the convolution sum, only $k = 0, 1, 2, 3$ survive:

$$y[n] = x[n] + x[n-1] - x[n-2] - x[n-3] = \underbrace{\big(x[n] + x[n-1]\big)}_{\text{recent pair}} - \underbrace{\big(x[n-2] + x[n-3]\big)}_{\text{older pair}}$$

The output is the difference between the two most recent inputs and the two before them, which measures how much the input changed. *(Random-input plot: output standard deviation about 2.)*

Step 3: Step response, $x[n] = u[n]$.

Running sum of the taps:

n	≤ -1	0	1	2	≥ 3
running sum	0	1	$1 + 1 = 2$	$2 - 1 = 1$	$1 - 1 = 0$

$$s[n] = \delta[n] + 2\delta[n-1] + \delta[n-2]$$

The final value is again 0, because the taps add up to 0. *(Step-response plot: values $1, 2, 1$, then 0.)*

Step 4: Frequency response, written out term by term.

$$H(e^{j\omega}) = 1 + e^{-j\omega} - e^{-j2\omega} - e^{-j3\omega}$$

Use $e^{-j\theta} = \cos\theta - j\sin\theta$ on every term:

$$\mathrm{Re}\{H\} = 1 + \cos\omega - \cos 2\omega - \cos 3\omega, \qquad \mathrm{Im}\{H\} = -\sin\omega + \sin 2\omega + \sin 3\omega$$

Step 5: Closed form by factoring (explains the shape of the plots).

Group in pairs and pull out the common factor:

$$H(e^{j\omega}) = \left(1 + e^{-j\omega}\right) - e^{-j2\omega}\left(1 + e^{-j\omega}\right) = \left(1 + e^{-j\omega}\right)\left(1 - e^{-j2\omega}\right)$$

The second factor is a difference of squares, $1 - a^2 = (1 - a)(1 + a)$ with $a = e^{-j\omega}$:

$$H(e^{j\omega}) = \left(1 - e^{-j\omega}\right)\left(1 + e^{-j\omega}\right)^2$$

Simplify each factor:

$$1 - e^{-j\omega} = 2j \sin(\omega/2)\, e^{-j\omega/2}$$
$$\left(1 + e^{-j\omega}\right)^2 = \left[2\cos(\omega/2)\, e^{-j\omega/2}\right]^2 = 4\cos^2(\omega/2)\, e^{-j\omega}$$

Multiply them back together:

$$\boxed{H(e^{j\omega}) = 8j\, \sin(\omega/2)\cos^2(\omega/2)\, e^{-j3\omega/2}}$$

$\sin(\omega/2)$ gives a single zero at $\omega = 0$, and $\cos^2(\omega/2)$ gives a double zero at $\omega = \pi$.

Step 6: Magnitude, phase, real, and imaginary parts.

$$H(e^{j\omega}) = \underbrace{8\sin(\omega/2)\cos^2(\omega/2)}_{\text{real amplitude}}\, e^{j\left(\frac{\pi}{2} - \frac{3\omega}{2}\right)}$$

$$|H(e^{j\omega})| = 8\,|\sin(\omega/2)|\cos^2(\omega/2)$$

$$\angle H(e^{j\omega}) = \begin{cases} \dfrac{\pi}{2} - \dfrac{3\omega}{2}, & \sin(\omega/2) > 0 \\ \dfrac{\pi}{2} - \dfrac{3\omega}{2} + \pi, & \sin(\omega/2) < 0 \end{cases} \qquad (\text{then wrapped into } (-\pi, \pi])$$

Since $\cos^2(\omega/2) \geq 0$, only $\sin(\omega/2)$ can change the sign. So the phase jumps by π at $\omega = 0$ but only wraps at $\omega = \pi$.

Peak of the magnitude. Let $s = \sin(\omega/2)$, so $\cos^2(\omega/2) = 1 - s^2$ and $|H| = 8s(1 - s^2)$ for $s \geq 0$. Setting the derivative to zero:

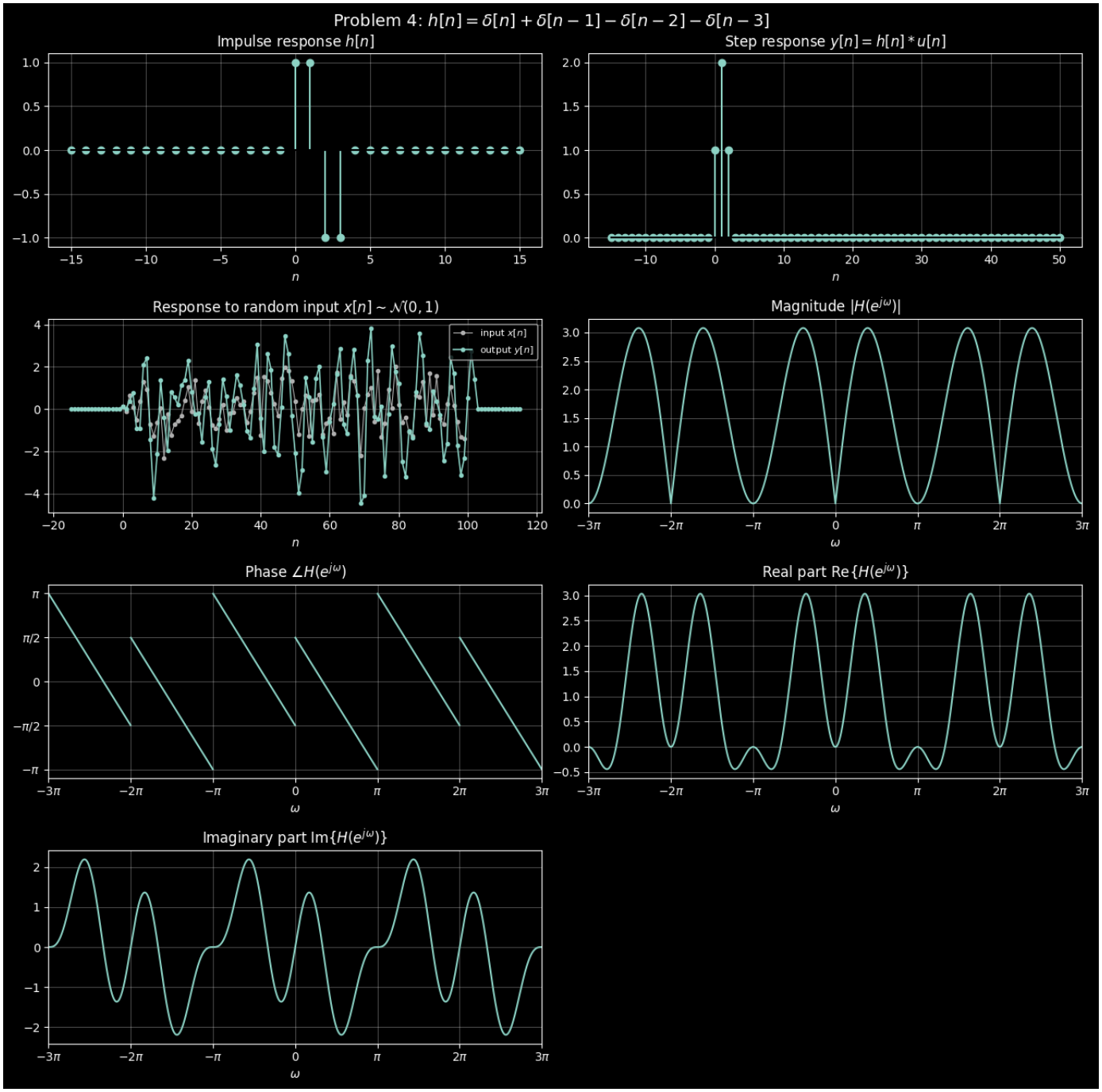
$$\frac{d}{ds}\left[8s - 8s^3\right] = 8 - 24s^2 = 0 \;\Rightarrow\; s = \frac{1}{\sqrt{3}} \;\Rightarrow\; |H|_{\max} = 8 \cdot \frac{1}{\sqrt{3}} \cdot \frac{2}{3} = \frac{16}{3\sqrt{3}} \approx 3.08$$

at $\omega = 2\arcsin(1/\sqrt{3}) \approx 0.39\pi$.

Step 7: Key values to check against the plots.

Quantity	Value	Where to see it
$H(e^{j0})$	$1+1-1-1$ $=0$	Sharp V-shaped zero in the magnitude at DC
$H(e^{j\pi})$	$1-1-1+1$ $=0$	Rounded, flat-bottomed zero at $\omega=\pm\pi$ (double zero)
Peak	$\lvert H\rvert_{\max}\approx 3.08$ at $\omega\approx\pm 0.39\pi$	Two peaks per period in the magnitude plot
Phase slope	$-3/2$ (1.5-sample delay)	Descending segments in the phase plot
Period	2π	Every frequency plot repeats at $\omega=\pm 2\pi$

```
ba check: 0.0
```



Classification

Property	Answer	Justification
Filter type	Bandpass	When I compare $\lvert H(e^{j\omega})\rvert$ at ω = 0 (DC) and ω = π, ω = π and ω = 0 are both maximally small with the peak at an angle somewhere in between
Causality	Causal	$h[n] = 0$ for all n < 0
Stability	Stable	$\Sigma\lvert h[n]\rvert = 4 < \infty$

Property	Answer	Justification
FIR or IIR	FIR	There are only 4 nonzero samples

Problem 5: $h[n] = 0.8^n\, u[n]$

Derivation

Step 1: Write out $h[n]$ sample by sample.

$u[n]$ turns the sequence on at $n = 0$. Each sample is 0.8 times the one before it:

n	≤ -1	0	1	2	3	4	5	$\cdots$
$h[n] = 0.8^n$	0	1	0.8	0.64	0.512	0.410	0.328	$\to 0$

The response decays toward zero but never reaches it, so $h[n]$ is infinitely long. *(Impulse-response plot.)*

Step 2: Output for any input $x[n]$, as a recursion.

From the convolution sum, with $h[k] = 0.8^k$ for $k \geq 0$:

$$y[n] = \sum_{k=0}^{\infty} 0.8^k\, x[n-k] = x[n] + \sum_{k=1}^{\infty} 0.8^k\, x[n-k]$$

In the remaining sum, factor out one 0.8 and substitute $k' = k - 1$:

$$\sum_{k=1}^{\infty} 0.8^k\, x[n-k] = 0.8 \sum_{k'=0}^{\infty} 0.8^{k'}\, x[(n-1)-k'] = 0.8\, y[n-1]$$

So the infinite sum collapses to a one-line recursion:

$$\boxed{y[n] = 0.8\, y[n-1] + x[n]}$$

Each output is the new input plus 80% of the previous output, which averages the input over time. *(Random-input plot: a smoothed input. After the input stops at $n = 99$, the output decays by a factor of 0.8 each step instead of stopping.)*

Step 3: Step response, $x[n] = u[n]$.

$s[n] = \sum_{k=0}^{n} h[k]$ is a finite geometric series with ratio 0.8:

$$s[n] = \sum_{k=0}^{n} 0.8^k = \frac{1 - 0.8^{n+1}}{1 - 0.8} = 5\left(1 - 0.8^{n+1}\right), \quad n \geq 0$$

n	0	1	2	3	$\cdots$	∞
$s[n]$	1	1.8	2.44	2.95	$\cdots$	5

(Step-response plot: rises smoothly and settles at 5.)

Step 4: Frequency response.

$$H(e^{j\omega}) = \sum_{n=0}^{\infty} 0.8^n e^{-j\omega n} = \sum_{n=0}^{\infty} \left(0.8\, e^{-j\omega}\right)^n$$

This is an infinite geometric series with ratio $a = 0.8\, e^{-j\omega}$. Since $|a| = 0.8 < 1$, it converges to $\frac{1}{1-a}$:

$$\boxed{H(e^{j\omega}) = \frac{1}{1 - 0.8\, e^{-j\omega}}}$$

Step 5: Separate the real and imaginary parts.

Expand the denominator with $e^{-j\omega} = \cos\omega - j\sin\omega$:

$$1 - 0.8\, e^{-j\omega} = (1 - 0.8\cos\omega) + j\, 0.8\sin\omega$$

Multiply the top and bottom by the conjugate, $(1 - 0.8\cos\omega) - j\, 0.8\sin\omega$. The denominator becomes real:

$$(1 - 0.8\cos\omega)^2 + (0.8\sin\omega)^2 = 1 - 1.6\cos\omega + 0.64\left(\cos^2\omega + \sin^2\omega\right) = 1.64 - 1.6\cos\omega$$

$$H(e^{j\omega}) = \frac{(1 - 0.8\cos\omega) - j\, 0.8\sin\omega}{1.64 - 1.6\cos\omega}$$

Step 6: Magnitude, phase, real, and imaginary parts.

$$\mathrm{Re}\{H\} = \frac{1 - 0.8\cos\omega}{1.64 - 1.6\cos\omega}, \qquad \mathrm{Im}\{H\} = \frac{-0.8\sin\omega}{1.64 - 1.6\cos\omega}$$

$$|H(e^{j\omega})| = \frac{1}{|1 - 0.8e^{-j\omega}|} = \frac{1}{\sqrt{1.64 - 1.6\cos\omega}},$$

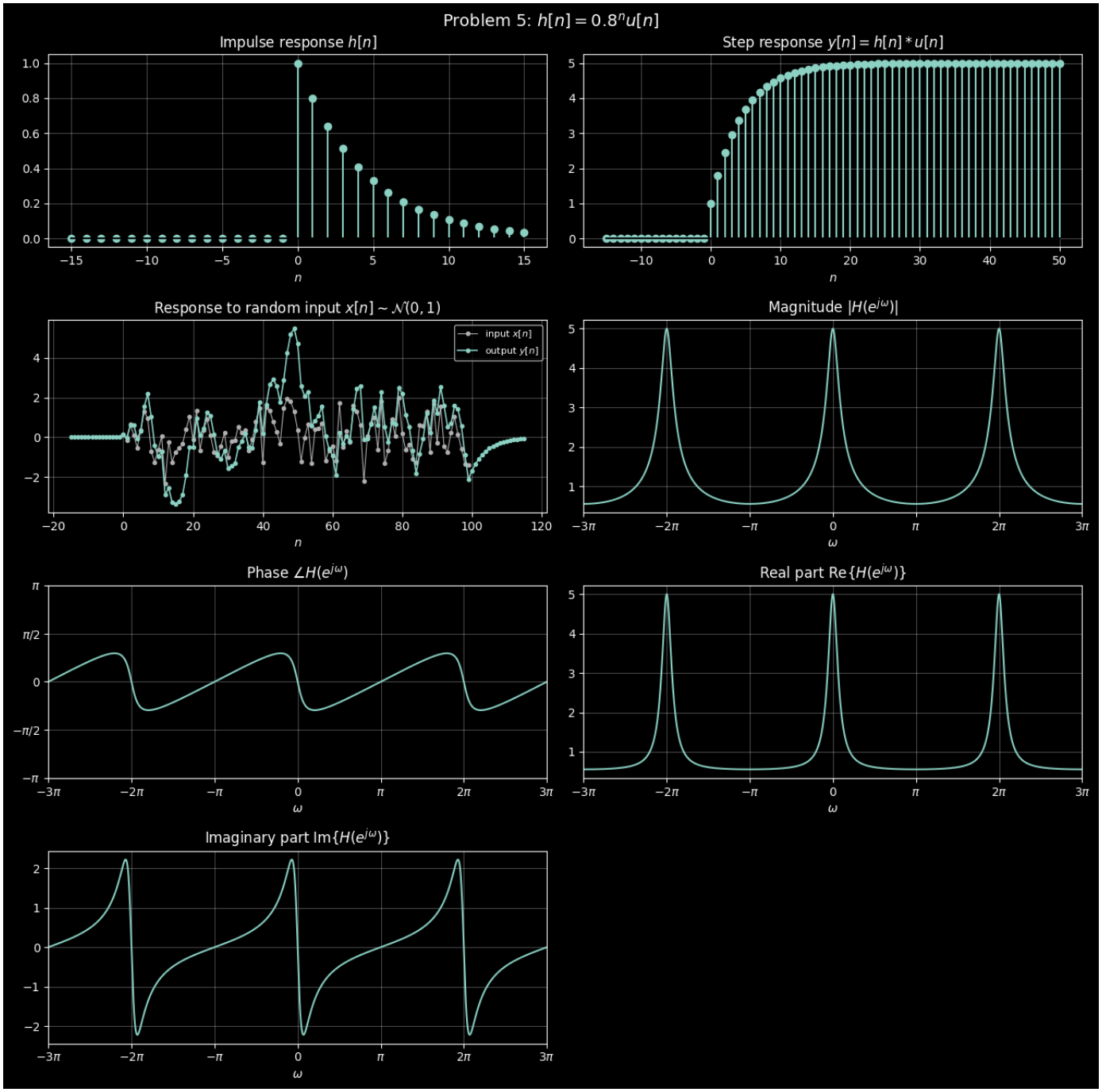
$$\angle H(e^{j\omega}) = -\arctan\left(\frac{0.8\sin\omega}{1 - 0.8\cos\omega}\right)$$

Since $1 - 0.8\cos\omega \geq 0.2 > 0$, the real part is always positive. The phase therefore never reaches $\pm\pi/2$ and needs no wrapping.

Step 7: Key values to check against the plots.

Quantity	Value	Where to see it
$H(e^{j0})$	$\dfrac{1}{1-0.8} = 5$	Peak of magnitude and real part; final value of the step response
$H(e^{j\pi})$	$\dfrac{1}{1+0.8} \approx 0.556$	Minimum of magnitude and real part at $\omega = \pm\pi$
Phase extremes	$\pm\arctan(4/3) = \pm 0.295\pi$ at $\omega \approx \mp 0.205\pi$	Phase plot; it never wraps
$\mathrm{Im}\{H\}$ extremes	± 2.22 at $\omega \approx \mp 0.07\pi$	Sharp swing in the imaginary part around DC
Pole	$z = 0.8$, inside the unit circle	Why the geometric series converges
Period	2π	Every frequency plot repeats at $\omega = \pm 2\pi$

```
ba check: 5.551115123125783e-17
```



Classification

Property	Answer	Justification
Filter type	Lowpass	When I compare $\lvert H(e^{j\omega})\rvert$ at ω = 0 (DC) and ω = π, the peak is at ω = 0 and most minimum at ω = π
Causality	Causal	$h[n] = 0$ for all n < 0
Stability	Stable	$\Sigma \lvert h[n] \rvert = 5 < \infty$
FIR or IIR	IIR	There are an infinite number of nonzero samples

Problem 6: $h[n] = (-0.8)^n\, u[n]$

Derivation

Step 1: Write out $h[n]$ sample by sample.

Each sample is -0.8 times the one before it, so the sign alternates while the size decays:

n	≤ -1	0	1	2	3	4	5	$\cdots$
$h[n] = (-0.8)^n$	0	1	-0.8	0.64	-0.512	0.410	-0.328	$\to 0$

This equals Problem 5's $h[n]$ multiplied by $(-1)^n$. *(Impulse-response plot.)*

Step 2: Output for any input $x[n]$, as a recursion.

The same steps as Problem 5, with 0.8 replaced by -0.8:

$$y[n] = \sum_{k=0}^{\infty} (-0.8)^k\, x[n-k] = x[n] + (-0.8) \sum_{k'=0}^{\infty} (-0.8)^{k'}\, x[(n-1)-k']$$

$$\boxed{y[n] = -0.8\, y[n-1] + x[n]}$$

Each output subtracts 80% of the previous output, so consecutive outputs push in opposite directions. *(Random-input plot: a jagged output that flips sign often. After $n = 99$ it decays while alternating sign.)*

Step 3: Step response, $x[n] = u[n]$.

A finite geometric series with ratio -0.8:

$$s[n] = \sum_{k=0}^{n} (-0.8)^k = \frac{1 - (-0.8)^{n+1}}{1 - (-0.8)} = \frac{5}{9}\left(1 - (-0.8)^{n+1}\right), \quad n \geq 0$$

n	0	1	2	3	4	$\cdots$	∞
$s[n]$	1	0.2	0.84	0.328	0.738	$\cdots$	$5/9 \approx 0.556$

(Step-response plot: overshoots and undershoots alternately, then settles at 0.556.)

Step 4: Frequency response.

An infinite geometric series with ratio $a = -0.8\, e^{-j\omega}$, where $|a| = 0.8 < 1$:

$$H(e^{j\omega}) = \sum_{n=0}^{\infty} \left(-0.8\, e^{-j\omega}\right)^n = \boxed{\frac{1}{1 + 0.8\, e^{-j\omega}}}$$

Step 5: Separate the real and imaginary parts.

$$1 + 0.8\, e^{-j\omega} = (1 + 0.8\cos\omega) - j\, 0.8\sin\omega$$

Multiply the top and bottom by the conjugate, $(1 + 0.8\cos\omega) + j\, 0.8\sin\omega$:

$$(1 + 0.8\cos\omega)^2 + (0.8\sin\omega)^2 = 1.64 + 1.6\cos\omega$$

$$H(e^{j\omega}) = \frac{(1 + 0.8\cos\omega) + j\, 0.8\sin\omega}{1.64 + 1.6\cos\omega}$$

Step 6: Magnitude, phase, real, and imaginary parts.

$$\mathrm{Re}\{H\} = \frac{1 + 0.8\cos\omega}{1.64 + 1.6\cos\omega}, \qquad \mathrm{Im}\{H\} = \frac{0.8\sin\omega}{1.64 + 1.6\cos\omega}$$

$$|H(e^{j\omega})| = \frac{1}{\sqrt{1.64 + 1.6\cos\omega}}, \qquad \angle H(e^{j\omega}) = \arctan\left(\frac{0.8\sin\omega}{1 + 0.8\cos\omega}\right)$$

Connection to Problem 5. Since $(-1)^n = e^{j\pi n}$, we have $h_6[n] = e^{j\pi n}\, h_5[n]$. Multiplying by $e^{j\pi n}$ shifts the spectrum by π:

$$H_6(e^{j\omega}) = \sum_n h_5[n]\, e^{j\pi n} e^{-j\omega n} = \sum_n h_5[n]\, e^{-j(\omega-\pi)n} = H_5\left(e^{j(\omega-\pi)}\right)$$

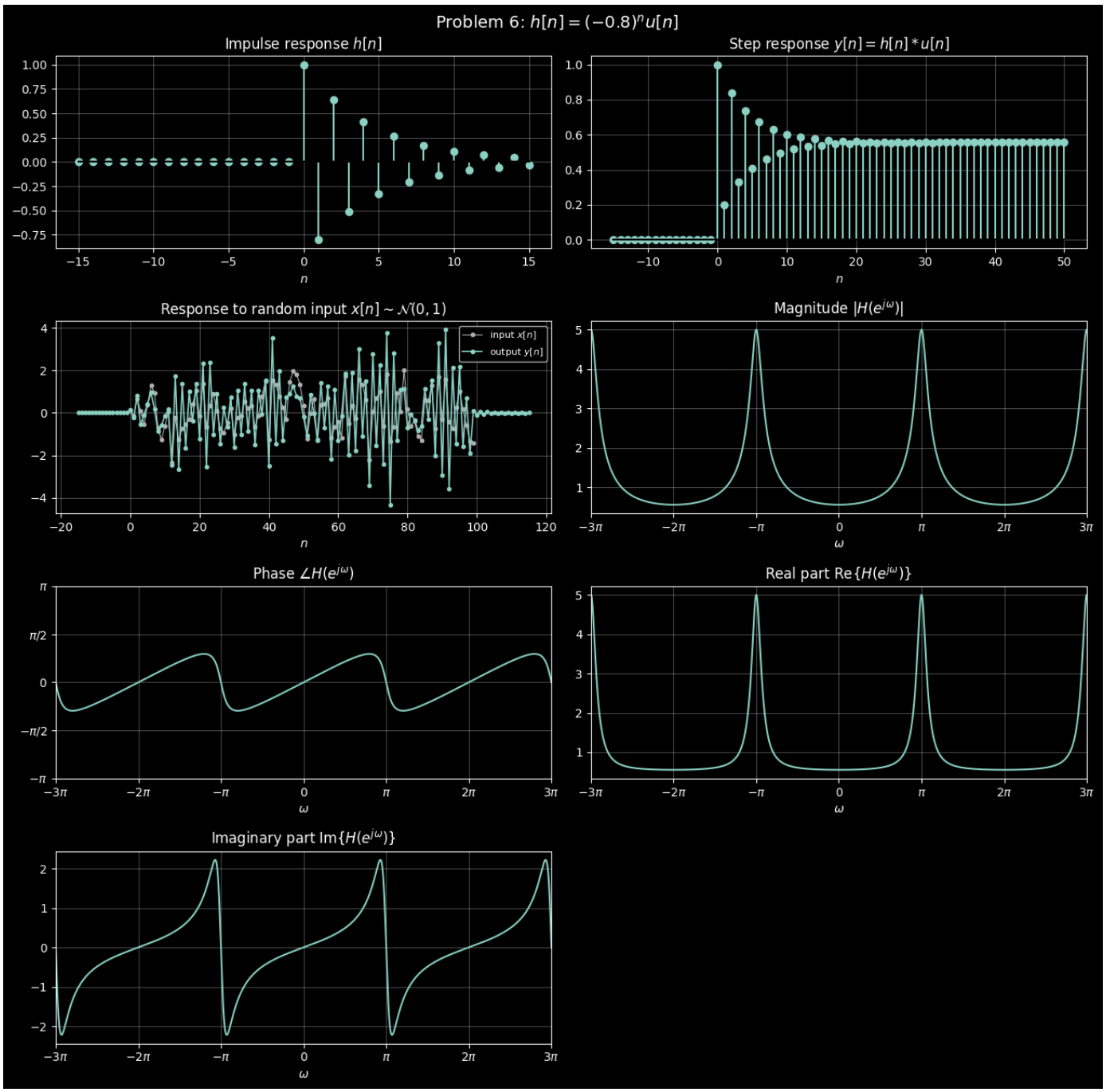
Every frequency plot is Problem 5's, slid over by π.

Step 7: Key values to check against the plots.

Quantity	Value	Where to see it
$H(e^{j0})$	$\dfrac{1}{1+0.8} \approx 0.556$	Minimum of magnitude at DC; final value of the step response
$H(e^{j\pi})$	$\dfrac{1}{1-0.8} = 5$	Peak of magnitude and real part at $\omega = \pm\pi$
Phase extremes	$\pm 0.295\pi$ at $\omega \approx \pm 0.795\pi$	Just before and after the steep drop through $\omega = \pm\pi$
$\mathrm{Im}\{H\}$ extremes	± 2.22 at $\omega \approx \pm 0.93\pi$	Sharp swing in the imaginary part around $\pm\pi$
Pole	$z = -0.8$, inside the unit circle	Why the geometric series converges

Quantity	Value	Where to see it
Period	2π	Every frequency plot repeats at $\omega = \pm 2\pi$

```
ba check: 5.551115123125783e-17
```



Classification

Property	Answer	Justification
Filter type	Highpass	When I compare $\lvert H(e^{j\omega})\rvert$ at ω = 0 (DC) and ω = π, the peak is at ω = π and most minimum at ω = 0

Property	Answer	Justification
Causality	Causal	$h[n] = 0$ for all n < 0
Stability	Stable	$\Sigma\lvert(-0.8)^n\rvert = \sum_n 0.8^n = 5 < \infty$
FIR or IIR	IIR	There are an infinite number of nonzero samples

Problem 7: $h[n] = 0.5\,(0.9^n)\cos(0.3\pi n + 0.9\pi)\,u[n]$

Derivation

Constants: $r = 0.9$, $\omega_0 = 0.3\pi$, $\phi = 0.9\pi$, so $h[n] = 0.5\,r^n\cos(\omega_0 n + \phi)\,u[n]$.

Step 1: Write out $h[n]$ sample by sample.

n	≤ -1	0	1	2	3	4	5	6	7	$\cdots$
$h[n]$	0	-0.476	-0.364	0	0.295	0.312	0.091	-0.156	-0.239	$\to 0$

The cosine has period $2\pi/(0.3\pi) = 6.\overline{6}$ samples, and the factor 0.9^n shrinks it. The sample at $n = 2$ is exactly 0 because $\cos(0.6\pi + 0.9\pi) = \cos(1.5\pi) = 0$. *(Impulse-response plot.)*

Step 2: Rewrite the cosine with Euler's formula.

Use $\cos\theta = \dfrac{e^{j\theta} + e^{-j\theta}}{2}$ with $\theta = \omega_0 n + \phi$:

$$h[n] = \frac{0.5\,r^n}{2}\left[e^{j(\omega_0 n+\phi)} + e^{-j(\omega_0 n+\phi)}\right]u[n] = \frac{1}{4}\left[e^{j\phi}\left(re^{j\omega_0}\right)^n + e^{-j\phi}\left(re^{-j\omega_0}\right)^n\right]u[n]$$

Name the complex pole $p = re^{j\omega_0} = 0.9\,e^{j0.3\pi}$, with conjugate $p^* = 0.9\,e^{-j0.3\pi}$:

$$h[n] = \frac{1}{4}\left[e^{j\phi}\,p^n + e^{-j\phi}\,(p^*)^n\right]u[n]$$

The damped cosine is now a sum of two geometric sequences, each of which can be summed in closed form.

Step 3: Frequency response.

Each term is an infinite geometric series with ratio $p\,e^{-j\omega}$ or $p^*e^{-j\omega}$. Both have magnitude $0.9 < 1$, so both converge:

$$H(e^{j\omega}) = \frac{1}{4}e^{j\phi}\sum_{n=0}^{\infty}\left(p\,e^{-j\omega}\right)^n + \frac{1}{4}e^{-j\phi}\sum_{n=0}^{\infty}\left(p^*e^{-j\omega}\right)^n = \frac{\frac{1}{4}e^{j\phi}}{1 - p\,e^{-j\omega}} + \frac{\frac{1}{4}e^{-j\phi}}{1 - p^*e^{-j\omega}}$$

Combine over a common denominator. Since $p + p^* = 2r\cos\omega_0$ and $pp^* = r^2$, the denominator is

$$\left(1 - p\,e^{-j\omega}\right)\left(1 - p^*e^{-j\omega}\right) = 1 - 2r\cos\omega_0\,e^{-j\omega} + r^2e^{-j2\omega}$$

and the numerator, after pairing the conjugate terms with Euler's formula, is

$$\tfrac{1}{4}\left[e^{j\phi}\left(1 - p^*e^{-j\omega}\right) + e^{-j\phi}\left(1 - p\,e^{-j\omega}\right)\right] = \tfrac{1}{2}\left[\cos\phi - r\cos(\phi - \omega_0)\,e^{-j\omega}\right]$$

Plug in the numbers: $\cos(0.9\pi) = -0.9511$, $r\cos(0.6\pi) = -0.2781$, $2r\cos(0.3\pi) = 1.0580$, and $r^2 = 0.81$:

$$\boxed{H(e^{j\omega}) = \frac{-0.4755 + 0.1391\,e^{-j\omega}}{1 - 1.0580\,e^{-j\omega} + 0.81\,e^{-j2\omega}}}$$

Step 4: Output for any input $x[n]$ (difference equation).

Write $H = B/A$ and cross-multiply, $A\,Y = B\,X$. Each $e^{-jk\omega}$ becomes a delay of k samples:

$$y[n] - 1.0580\,y[n-1] + 0.81\,y[n-2] = -0.4755\,x[n] + 0.1391\,x[n-1]$$

$$\boxed{y[n] = 1.0580\,y[n-1] - 0.81\,y[n-2] - 0.4755\,x[n] + 0.1391\,x[n-1]}$$

The two feedback terms make the output ring at $\omega_0 = 0.3\pi$. *(Random-input plot: the output looks like a noisy oscillation with a period of about 6.7 samples, and it keeps ringing after the input stops at $n = 99$.)*

Step 5: Step response, $x[n] = u[n]$.

Since $h[n] = \mathrm{Re}\{0.5\,e^{j\phi}p^n\}$ for $n \geq 0$, the running sum is the real part of a finite geometric series:

$$s[n] = \sum_{k=0}^{n} h[k] = \mathrm{Re}\left\{0.5\,e^{j\phi}\sum_{k=0}^{n}p^k\right\} = \mathrm{Re}\left\{\frac{0.5\,e^{j\phi}}{1-p}\left(1 - p^{\,n+1}\right)\right\}$$

With $C = \dfrac{0.5\,e^{j\phi}}{1-p} = 0.577\,e^{-j0.783\pi}$, so that $\mathrm{Re}\{C\} = -0.447$:

$$s[n] = \underbrace{-0.447}_{\text{final value}} - \underbrace{0.577\,(0.9)^{n+1}\cos\big(0.3\pi(n+1) - 0.783\pi\big)}_{\text{decaying ring}}, \quad n \geq 0$$

(Step-response plot: it starts at $s[0] = h[0] = -0.476$, rings, and settles at -0.447.)

Step 6: Magnitude, phase, real, and imaginary parts.

With $H = B/A$:

$$|H| = \frac{|B|}{|A|}, \qquad \angle H = \angle B - \angle A, \qquad \mathrm{Re}\{H\} = \frac{\mathrm{Re}\{BA^*\}}{|A|^2}, \qquad \mathrm{Im}\{H\} = \frac{\mathrm{Im}\{BA^*\}}{|A|^2}$$

Why the peaks are at $\pm 0.3\pi$. The denominator factors into two distances:

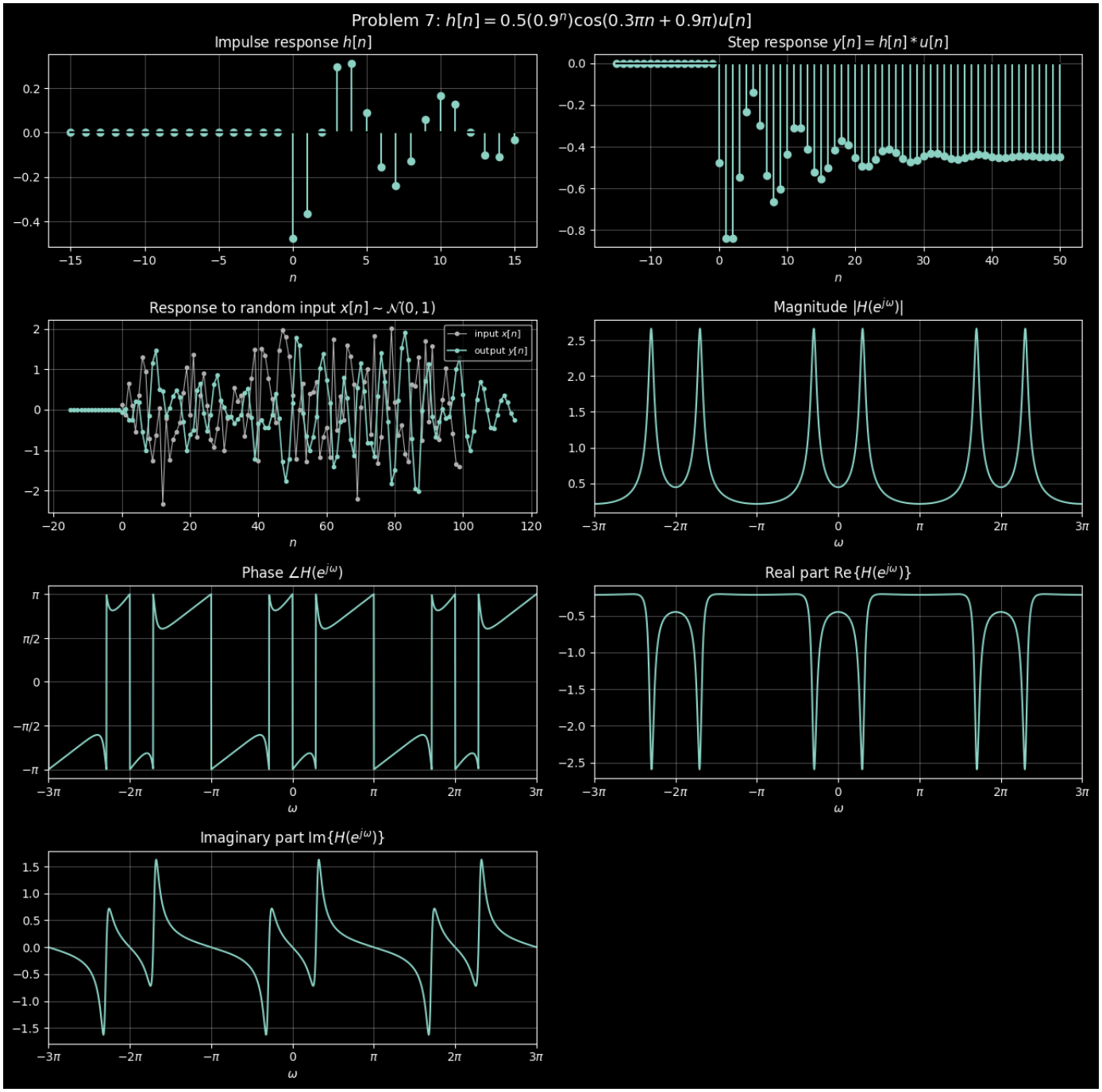
$$|A| = \left|1 - p\,e^{-j\omega}\right| \left|1 - p^* e^{-j\omega}\right| = \left|e^{j\omega} - p\right| \left|e^{j\omega} - p^*\right|$$

Each factor is the distance from the point $e^{j\omega}$ on the unit circle to a pole. When $\omega = 0.3\pi$, the point $e^{j\omega}$ is directly next to $p = 0.9\,e^{j0.3\pi}$, a distance of only $1 - 0.9 = 0.1$. A small denominator gives a large $|H|$. Moving the poles closer to the unit circle would make the peak taller and narrower.

Step 7: Key values to check against the plots.

Quantity	Value	Where to see it
$H(e^{j0})$	$\dfrac{-0.4755 + 0.1391}{1 - 1.0580 + 0.81} = -0.447$	Dip between the two peaks; final value of the step response
$H(e^{j\pi})$	$\dfrac{-0.4755 - 0.1391}{1 + 1.0580 + 0.81} = -0.214$	Floor of the magnitude at $\omega = \pm\pi$
Peak	$H(e^{j0.3\pi}) \approx -2.53 + 0.82j = 2.66\,e^{j0.90\pi}$	Magnitude peaks at $\pm 0.3\pi$
$\mathrm{Re}\{H\}$	Negative for every ω; minimum ≈ -2.59 near $\pm 0.3\pi$	Real-part plot; keeps the phase near $\pm\pi$
$\mathrm{Im}\{H\}$ extremes	$+1.63$ at 0.324π, -0.72 at 0.255π (mirrored for $\omega < 0$)	Imaginary-part plot
Poles	$p = 0.9\,e^{\pm j0.3\pi} = 0.529 \pm 0.728j$, inside the unit circle	Why the series converges and where the peaks are
Period	2π	Every frequency plot repeats at $\omega = \pm 2\pi$

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ba check: 1.942890293094024e-16
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Classification

Property	Answer	Justification
Filter type	Bandpass	When I compare $\left\|H(e^{j\omega})\right\|$ at ω = 0 (DC) and ω = π, ω = π and ω = 0 are both small with the peak at an angle somewhere in between
Causality	Causal	$h[n] = 0$ for all n < 0

Property	Answer	Justification
Stability	Stable	$\lvert h[n]\rvert \le 0.5\,(0.9)^n$, so $\Sigma\lvert h[n]\rvert \le \frac{0.5}{1-0.9} = 5 < \infty$. The poles $0.9e^{\pm j0.3\pi}$ are inside the unit circle.
FIR or IIR	IIR	There are an infinite number of nonzero samples

Problem 8: $h[n] = 1$

Derivation

Step 1: Write out $h[n]$ sample by sample.

n	$\cdots$	-2	-1	0	1	2	$\cdots$
$h[n]$	1	1	1	1	1	1	1

It's 1 for every n, forever in both directions. *(Impulse-response plot.)*

Step 2: Output for any input $x[n]$.

In the convolution sum every term has weight 1:

$$y[n] = \sum_{k=-\infty}^{\infty} (1)\, x[n-k]$$

Substitute $m = n - k$. As k runs over all integers, so does m:

$$\boxed{y[n] = \sum_{m=-\infty}^{\infty} x[m]}$$

The right side doesn't depend on n, so every output sample equals the sum of the entire input. *(Random-input plot: a flat line at $\sum_{m=0}^{99} x[m] \approx 8.11$ for this seed. A sum of 100 unit-variance samples has standard deviation 10, so 8 is typical.)*

Step 3: Step response, $x[n] = u[n]$.

$$s[n] = \sum_{k=-\infty}^{n} h[k] = \sum_{k=-\infty}^{n} 1 = \infty$$

The step response is infinite for every n: a bounded input produces an unbounded output.

What the plot shows instead. The code builds the step input only on $n \le 99$, which is 100 ones. With a finite input, Step 2 gives $y[n] = \sum_{m=0}^{99} 1 = 100$, which is why the plot shows a flat line at 100. A longer input window would give a larger number.

Step 4: Frequency response.

$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} e^{-j\omega n}$$

Every term has magnitude 1, so the terms never shrink and the sum does not converge as an ordinary sum. To get something we can plot, truncate to $|n| \le N$. This is exactly Problem 2's sum with 3 replaced by N, so the same steps (pair $\pm n$, or re-index and use the geometric series) give

$$H_N(e^{j\omega}) = \sum_{n=-N}^{N} e^{-j\omega n} = 1 + 2\sum_{n=1}^{N} \cos(\omega n) = \boxed{\frac{\sin\big((2N+1)\omega/2\big)}{\sin(\omega/2)}}$$

The plots use $N = 200$, which is 401 terms.

Step 5: What happens as $N \to \infty$.

H_N is a Dirichlet kernel, the same shape as Problem 2's but with 401 terms instead of 7:

- **Height at $\omega = 0$:** $H_N(e^{j0}) = 2N + 1 = 401$, growing without bound.
- **Width:** the first zeros are at $\omega = \pm\dfrac{2\pi}{2N+1} \approx \pm 0.016$, shrinking to 0.
- **Area over one period:** $\displaystyle\int_{-\pi}^{\pi} H_N(e^{j\omega})\, d\omega = \sum_{n=-N}^{N} \int_{-\pi}^{\pi} e^{-j\omega n}\, d\omega = 2\pi$, because only the $n = 0$ term survives the integral. The area stays 2π for every N.

A spike that grows taller and narrower while keeping area 2π approaches $2\pi\,\delta(\omega)$. Repeating it every 2π:

$$\boxed{H(e^{j\omega}) = 2\pi \sum_{k=-\infty}^{\infty} \delta(\omega - 2\pi k)}$$

Step 6: Magnitude, phase, real, and imaginary parts (truncated version).

H_N is real because $h[n]$ is even, so

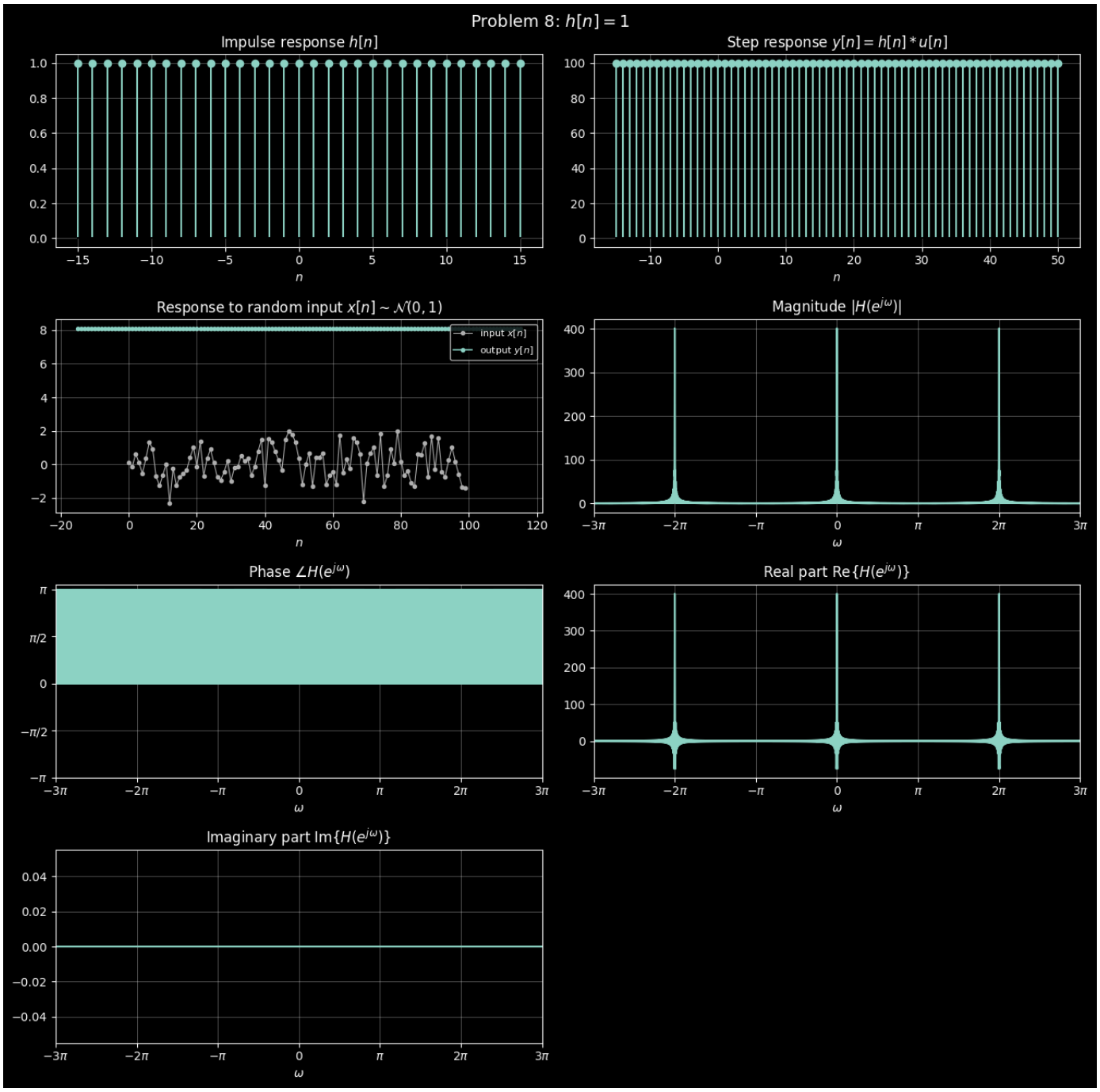
$$\mathrm{Re}\{H_N\} = \frac{\sin(401\,\omega/2)}{\sin(\omega/2)}, \qquad \mathrm{Im}\{H_N\} = 0, \qquad |H_N| = \left|\frac{\sin(401\,\omega/2)}{\sin(\omega/2)}\right|$$

The phase is 0 where $H_N > 0$ and π where $H_N < 0$. H_N changes sign at every zero, $\omega = 2\pi k/401$, which is about every 0.016 rad. That's faster than the plot's frequency grid, so the

phase line fills in as a solid band between 0 and π. For the true system, $H = 0$ away from the impulses, so its phase is undefined there.

Step 7: Key values to check against the plots.

Quantity	Value	Where to see it
Spike height	$2N + 1 = 401$ at $\omega = 0, \pm 2\pi$	Magnitude and real-part plots
Spike area	2π (for every N)	Why the spikes approach $2\pi\,\delta(\omega)$
Zero spacing	$2\pi/401 \approx 0.016$ rad	Solid band in the phase plot
$\mathrm{Im}\{H\}$	Exactly 0	Flat imaginary-part plot
Step response	∞ (plotted as 100 because of the 100-sample input window)	Step-response plot
Random-input output	$\sum x[m] \approx 8.11$, constant in n	Flat line in the random-input plot


Problem 8: $h[n] = 1$
Impulse response $h[n]$
Step response $y[n] = h[n] * u[n]$
Response to random input $x[n] \sim \mathcal{N}(0, 1)$
input $x[n]$
output $y[n]$
Magnitude $|H(e^{j\omega})|$
Phase $\angle H(e^{j\omega})$
Real part $\mathrm{Re}\{H(e^{j\omega})\}$
Imaginary part $\mathrm{Im}\{H(e^{j\omega})\}$
n
ω

Classification

Property	Answer	Justification
Filter type	Lowpass	$H(e^{j\omega})$ spikes at ω = 0 with $\lvert H(e^{j\omega})\rvert = 0$ when $\omega \neq 0$
Causality	Non-causal	There are nonzero samples when n < 0
Stability	Unstable	$\Sigma\lvert h[n]\rvert = \infty$
FIR or IIR	IIR	There are an infinite number of nonzero samples

Problem 9: $h[n] = 0.5\cos(0.3\pi n + 0.9\pi)$

Derivation

Constants: $\omega_0 = 0.3\pi$ and $\phi = 0.9\pi$, so $h[n] = 0.5\cos(\omega_0 n + \phi)$ for every n.

Step 1: Write out $h[n]$ sample by sample.

n	$\cdots$	-3	-2	-1	0	1	2	3	4	$\cdots$
$h[n]$	$\cdots$	0.5	0.294	-0.155	-0.476	-0.405	0	0.405	0.476	$\cdots$

A cosine of amplitude 0.5 and period $6.\overline{6}$ samples that never decays and extends in both directions. It's not symmetric about $n = 0$ (for example $h[1] \neq h[-1]$) because of the $\phi = 0.9\pi$ offset. *(Impulse-response plot.)*

Step 2: Rewrite the cosine with Euler's formula.

Use $\cos\theta = \dfrac{e^{j\theta} + e^{-j\theta}}{2}$:

$$h[n] = \frac{1}{4}\left[e^{j\phi}e^{j\omega_0 n} + e^{-j\phi}e^{-j\omega_0 n}\right]$$

Equivalently, $h[n] = 0.5\ \mathrm{Re}\{e^{j(\omega_0 n+\phi)}\}$.

Step 3: Output for any input $x[n]$.

Write the convolution with h shifted, $y[n] = \sum_k x[k]\,h[n-k]$, and use $\cos(\omega_0(n-k)+\phi) = \mathrm{Re}\{e^{j(\omega_0 n+\phi)}\,e^{-j\omega_0 k}\}$:

$$y[n] = \sum_k x[k] \cdot 0.5\cos\big(\omega_0(n-k)+\phi\big) = 0.5\ \mathrm{Re}\left\{ e^{j(\omega_0 n+\phi)} \underbrace{\sum_k x[k]\, e^{-j\omega_0 k}}_{X(e^{j\omega_0})} \right\}$$

The sum is the input's DTFT evaluated at the single frequency ω_0. Writing $X(e^{j\omega_0}) = |X|\, e^{j\angle X}$:

$$\boxed{y[n] = 0.5\,|X(e^{j\omega_0})|\cos\big(\omega_0 n + \phi + \angle X(e^{j\omega_0})\big)}$$

Whatever the input, the output is a pure sinusoid at $\omega_0 = 0.3\pi$ for every n. Only the input's content at ω_0 sets its amplitude and phase. *(Random-input plot: $|X(e^{j0.3\pi})| = 9.10$ for this seed, so the output is a steady sinusoid of amplitude 4.55 running across the whole plot, including before the input starts and after it ends.)*

Step 4: Step response, $x[n] = u[n]$.

$$s[n] = \sum_{k=-\infty}^{n} 0.5\cos(0.3\pi k + 0.9\pi)$$

The terms never shrink, so the partial sums keep oscillating and never settle: the step response does not converge.

What the plot shows instead. The code's step input is $L = 100$ ones ($n = 0, \ldots, 99$). Plug that into Step 3. The input's DTFT at ω_0 is a finite geometric series, and the same steps as Problem 2, Step 5 give

$$X(e^{j\omega_0}) = \sum_{m=0}^{L-1} e^{-j\omega_0 m} = e^{-j\omega_0(L-1)/2}\,\frac{\sin(L\omega_0/2)}{\sin(\omega_0/2)}$$

$$y[n] = 0.5\,\frac{\sin(L\omega_0/2)}{\sin(\omega_0/2)}\cos\left(\omega_0 n + \phi - \frac{\omega_0(L-1)}{2}\right)$$

With $L = 100$: $\sin(100 \cdot 0.15\pi) = \sin(15\pi) = 0$, so $y[n] = 0$ exactly. The input window covers exactly 15 periods of the cosine, and a cosine sums to zero over whole periods. The 10^{-14} values in the plot are round-off. With $L = 101$ the amplitude would be 0.50, and the largest possible is $0.5/\sin(0.15\pi) \approx 1.10$.

Step 5: Frequency response.

Truncate to $|n| \le N$ and apply Problem 8's result to each exponential from Step 2. Each one is Problem 8's sum shifted in frequency by $\pm\omega_0$:

$$H_N(e^{j\omega}) = \frac{1}{4}e^{j\phi}\sum_{n=-N}^{N} e^{-j(\omega-\omega_0)n} + \frac{1}{4}e^{-j\phi}\sum_{n=-N}^{N} e^{-j(\omega+\omega_0)n} = \frac{1}{4}e^{j\phi}\,D_N(\omega-\omega_0) + \frac{1}{4}e^{-j\phi}\,D_N(\omega+\omega_0)$$

where $D_N(\theta) = \dfrac{\sin\big((2N+1)\theta/2\big)}{\sin(\theta/2)}$ is Problem 8's Dirichlet kernel. As $N \to \infty$, each D_N becomes $2\pi\,\delta$:

$$\boxed{H(e^{j\omega}) = \frac{\pi}{2}\sum_{k=-\infty}^{\infty}\left[e^{j0.9\pi}\,\delta(\omega - 0.3\pi - 2\pi k) + e^{-j0.9\pi}\,\delta(\omega + 0.3\pi - 2\pi k)\right]}$$

Step 6: Magnitude, phase, real, and imaginary parts (truncated version, $N = 200$).

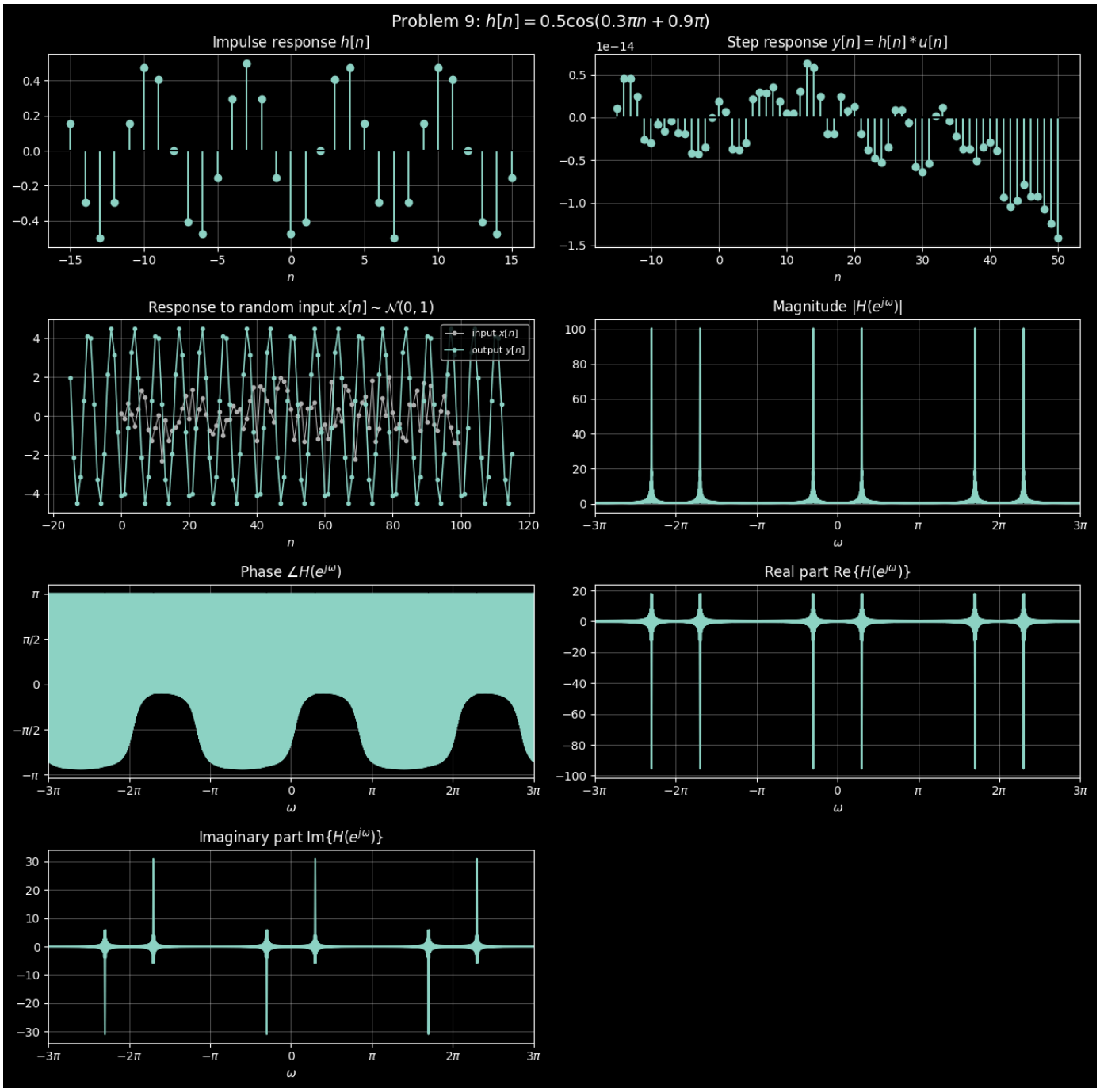
Near $\omega = +\omega_0$ the first kernel dominates, with peak $D_N(0) = 401$:

$$H_N(e^{j0.3\pi}) \approx \frac{401}{4}\,e^{j0.9\pi} = 100.25\,[\cos(0.9\pi) + j\sin(0.9\pi)] \approx -95.3 + 31.0j$$

Near $\omega = -\omega_0$ the second kernel dominates and gives the complex conjugate, $\approx -95.3 - 31.0j$. Away from $\pm\omega_0$, both kernels are tiny compared with these spikes. The phase is meaningful only at the spikes, where it equals $\pm\phi = \pm 0.9\pi$. Elsewhere the true H is 0 and its phase is undefined, so the plotted phase is a filled band, as in Problem 8.

Step 7: Key values to check against the plots.

Quantity	Value	Where to see it
Spike locations	$\omega = \pm 0.3\pi$ (and $\pm 1.7\pi$, $\pm 2.3\pi$ by periodicity)	Magnitude plot
Spike height	$\frac{401}{4} \approx 100$	Magnitude plot
Value at $+0.3\pi$	$\approx 100\,e^{j0.9\pi} = -95.3 + 31.0j$	Real part ≈ -95; imaginary part $\approx +31$
Value at -0.3π	$\approx -95.3 - 31.0j$	Real part ≈ -95; imaginary part ≈ -31
Impulse weights (true H)	$\frac{\pi}{2}e^{\pm j0.9\pi}$	Limit as $N \to \infty$
Step response	Does not converge (plotted as 0 because 100 samples is exactly 15 periods)	Step-response plot, 10^{-14} scale
Random-input output	Sinusoid at 0.3π, amplitude $0.5\,\lvert X(e^{j0.3\pi})\rvert = 4.55$	Random-input plot



Classification

Property	Answer	Justification
Filter type	Bandpass	$\lvert H(e^{j\omega})\rvert = 0$ when ω = 0 and ω = π and is ∞ when ω = 0.3π
Causality	Non-causal	There are nonzero samples when n < 0
Stability	Unstable	The sinusoidal in $h[n]$ makes $h[n]$ nonconvergent

Property	Answer	Justification
FIR or IIR	IIR	There are an infinite number of nonzero samples

What, if anything, did you learn from this homework?

Problems 1 & 2: A time shift changes the phase.

Problem 5 & 6: Multiplying an impulse response with $(-1)^n$ shifts the spectrum by π, making a lowpass into a highpass.

Problem 7: $|H(e^{j\omega})|$ is influenced by the location of the poles. The closer a pole is to the unit circle, the more $|H(e^{j\omega})|$ will peak.

What new questions, if any, do you have?

I have no questions at this time.